Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/30/2007
PI Name:	Dunn, Catherine Ph.D.		
Project Title:	How the brain constructs stable visual representations: cortical and subcortical mechanisms		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cad11@pitt.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	412-268-2010
Organization Name:	University of Pittsburgh		
PI Address 1:	Department of Neuroscience		
PI Address 2:	446 Crawford Hall		
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15213	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	08/01/2004	**End Date:**	07/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	When making our way through the environment our eyes are constantly moving, scanning the scene. Though images move on the retina, our perception of the world remains stable. What is the neural basis of this stability? Neurons in several cortical areas are capable of “remapping” visual representations in conjunction with eye movements. Recent experiments have suggested, surprisingly, that subcortical pathways can be recruited as well when the normal cortical pathways are interrupted. The aim of the proposed experiments is to determine how subcortical pathways contribute to solving the problem of maintaining stable spatial representations. The aim of the proposed experiments is to determine how subcortical pathways contribute to solving the problem of maintaining stable spatial representations. The human mind uses both external and internal sensory cues to create frames of reference for motor plans. In outer		

	space, the internal sensory cues are modified. Signals (from otolith organs) about acceleration and gravity are changed, causing a discrepancy in external visual signals and internal sensory systems (Clement, 1998; Pozzo, 1998). As a result, astronauts experience disorientation, motion sickness and problems with eye-hand coordination (Glasauer and Mittlstaedt, 1998; Bock, 1998). Astronauts become more dependent on their visual system to create a reliable framework. In order to understand how the system is modified in outer space, we first must understand how stable visual representations are formed in the normal environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	My work is focused on determining the brain circuits that contribute to creation of stable visual representations. These interests build upon illuminating studies done previously in the lab that focused on the transfer of visual information for the updating of an internal representation. We asked whether updating of a visual representation across hemispheres depends on the forebrain commissures, the fiber bundle that connects the cortical hemispheres. We tested the integrity of spatial updating in two ways. First, we used a behavioral test that requires spatial information to be updated when the eyes move. Second, we analyzed the neural signal in the Laterial Intraparietal cortex (LIP), an area known to contain remapping signals. In both sets of experiments we examined the difference between a condition where the internal representation of the visual stimulus remained within a single hemisphere and a second condition where the internal representation must pass across to the opposite hemisphere. We hypothesized that when the internal representation of a visual stimulus remains within a single hemisphere, the behavioral task would be performed correctly and LIP neurons would have a normal remapped response. In contrast, when the representation of the stimulus has to move across hemispheres, we expected that the monkeys would be unable to perform the task and LIP neurons would not have a remapped response. A first, the monkeys' performance confirmed our hypothesis. The monkeys were unable to complete the task when the visual information needed to pass across hemispheres. Surprisingly, recovery began almost at once. Both monkeys ultimately recovered to the point where they could correctly perform the task that required across-hemisphere remapping of visual information. Additionally, when we recorded physiological activity in area LIP, we found neurons that fired both when the representation remained within a hemisphere and when the representation crossed hemispheres. The goal of my study is to determine the brain mechanism involved in spatial updating in monkeys when the forebrain commissures have been transected. Over the past year I had made three important discoveries. First, I found that even in the absence of the forebrain commissures, neural activity in LIP is related to the monkey's behavior during a task dependent on accurate spatial updating. These results suggest that LIP still plays a role in the behavior of the monkeys. After the removal of the forebrain commissures, the behavioral recovery might have been due entirely to subcortical mechanisms. Instead, it appears that information about the opposite visual field still reaches LIP, and this activity contributes to the overall behavior of the monkey. Second, I determined that the remapping activity in the split brain animals is not due to representation of ipsilateral and contralateral visual fields in a single hemisphere. I found that while in normal monkeys there is a small representation of ipsilateral space, that represenatation is absent when the forebrain commissures have been transected. This result indicates that a sub-cortical structure must be the source of the across hemifield remapping observed in the split brain animals. Third, I determined that neurons in the superior colliculus (SC) in the intact monkey are capable of remapping when the visual information remains within or must pass across hemifields. The difference between the LIP and SC is the number of neurons that have remapping activity. In LIP, we found that approximately 70% of the neurons have remapping activity; this is compared to about 30% in the SC. Even thought the total numbers of neurons are different, theses structures are similar in their participation in across and within hemifield remapping. Neurons in both areas have potential access to visual information from large portions of the visual field. The across hemifield remapping observed in the SC may be the source of the across hemifield remapping that remains in the LIP of the split brain monkey. Our ability to act within our environment, whether it is here on earth, or in space, depends on a stable perception of the world. Over the past year, my works has contributed to our understanding of the circuitry behind internal representations that helps create these stable perceptions.
Bibliography Type:	Description: (Last Updated: 11/02/2007)
Articles in Peer-reviewed Journals	Berman RA, Heiser LM, Dunn CA, Saunders RC, Colby CL. "Dynamic circuitry for updating spatial representations. III. From neurons to behavior." J Neurophysiol. 2007 Jul;98(1):105-21. PMID: 17493922 , Jul-2007